EE1007
Introduction to Computer Science Laboratory

Lab 6
Functions

Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Lab 6.1
Debugging Practice: GCD Calculator

- In this problem, there are 8 errors hidden in different lines of the given .cpp file. Students are asked to fixed these errors based on the provided code.
- Note that, rewriting the whole code is **forbidden**. For each error at each line, students are allowed to rewrite the line where an error at or use one or two additional lines to fix the errors at each line.

Lab 6.1
Debugging Practice: GCD Calculator

- The program takes three numbers as inputs.
- Once the program receives the inputs, it will calculate the factorial of each input number.
- After that, the program will take the calculated factorial numbers to find the GCD of these three factorial numbers .



Lab 6.1
Debugging Practice: GCD Calculator

- Input:
  - Three **integers**
  - **The range of user-provided integer is [1, 2^{31}-1]**
- Output:
  - GCD of the three factorial numbers



Lab 6.1
Debugging Practice: GCD Calculator

◆The screen dialog should appear as follows:

```
Enter 3 nums: 1 4 3
GCD of 1 and 24 and 6 is 1
Process returned 0 (0x0)   execution time : 3.959 s
Press any key to continue.
```


Lab 6.1
Debugging Practice: GCD Calculator

◆錯誤的地方：

```
4  int gcd(int x, int y){
5      int a, b, r;
6      if(x > y){
7          a = x;
8          b = y;
9      }
10     else{
11         a = y;
12         b = x;
13     }
14     while(b != 0){
15         r = a / b;
16         a = b;
17         b = r;
18     }
19     return a
20 }
21
```

```
34 double fact(int a){
35     int n;
36     int c;
37
38     n = -1;
39     c = 0;
40
41     while(n<=a){
42         c = c * n;
43         n = n + 1;
44     }
45     return c
46 }
```

```
22 int main(){
23     int ain = 6, bin = 12, cin = 9;
24     cin >> a;
25     cin >> b;
26     cin >> c;
27     ain = fact(ain)
28     bin = fact(bin)
29     cin = fact(cin)
30     cout<< "GCD of " << ain << ", " << bin << ", and " << cin << " is " << gcd(ain, gcd(bin, cin) );
31     return 0
32 }
33
```



Lab 6.2
Area and Perimeter



- In this problem, you are asked to find the area and the perimeter of a rectangle based on the first and the second given numbers.
- The program takes the first two numbers as side length and width of the rectangle, and finds the corresponding values of the area and the perimeter **by functions**.
- Besides, you are also asked to find the area and perimeter of a triangle, based on all three given numbers as each side length of the triangle.
- The program takes these three numbers as each side length of the triangle, and finds the corresponding values of the area and the perimeter by functions.





Lab 6.2
Area and Perimeter



- A triangle is **valid if the sum of its two sides is greater than its third side and the difference of its two sides is less than its third side.**
- If the numbers can't construct a triangle, the program will output "error", and ask to input the numbers again.
- If the answer is not an integer, the program will **round it to the integer**.



Lab 6.2
Area and Perimeter

- Students **must** to use functions shown below to find the GCD of given numbers, otherwise you will get 0 point.
  - void rectangleArea(…);
  - void rectanglePerimeter(…);
  - void triangleArea(…);
  - void trianglePerimeter(…);

Lab 6.2
Area and Perimeter

- Input:
  - Three **integers**
  - **The range of user-provided integer is [1, 2^{31}-1]**
- Output:
  - If the answer is not an integer, the program will round it to the integer.
  - Four corresponding integer answers.

Lab 6.2
Area and Perimeter

- Hint:

  You can use Heron formula to calculate the area of triangle.

- Triangle Area= $\sqrt{s(s-a)(s-b)(s-c)}$

  $s=(a+b+c)/2$

Lab 6.2
Area and Perimeter

- The screen dialog should appear as follows:

```
Input 3 Numbers: 1 1 2
error
Input 3 Numbers: 4 4 5
Perimeter of the rectangle: 16
Area of the rectangle: 16
Perimeter of the triangle: 13
Area of the triangle: 8

Process returned 0 (0x0)   execution time : 8.831 s
Press any key to continue.
```

Lab 6.2
Area and Perimeter

◆思考方向：



Lab 6.2
Area and Perimeter

◆Step 1.呼叫void function，並在裡頭cout出結果：

```
7   //call by value
8   void rectangleArea(int in1, int in2);
9   void rectanglePerimeter(int in1, int in2);
10
11  //call by ref
12  void triangleArea(int &in1, int &in2, int &in3);
13  void triangleArea(int &in1, int &in2, int &in3);
```

```
38  //rectangle
39  void rectanglePerimeter(int in1, int in2){
40      cout << "Perimeter of the rectangle: " << in1 * 2 + in2 * 2 << endl;
41  }
42  void rectangleArea(int in1, int in2){
43      cout << "Area of the rectangle: " << in1 * in2 << endl;
44  }
45  //triangle
46  void trianglePerimeter(int &in1, int &in2, int &in3){
47      cout << "Perimeter of the triangle: " << in1 + in2 + in3 << endl;
48  }
49  void triangleArea(int &in1, int &in2, int &in3){
50      //area=sqrt((a+b+c)*(-a+b+c)*(a-b+c)*(a+b-c))/4
51      cout << "Area of the triangle: " << roundToInt( sqrt((in1+in2+in3)*(in2+in3-in1)*(in1-in2+in3)*(in1+in2-in3)) / 4.0 )<< endl;
52  }
```

Lab 6.2
Area and Perimeter

◆Step 2.設計四捨五入function：

```
54  //round to int
55  double roundToInt(double x){
56      double y = floor( x + 0.5) ;
57      return y;
58  }
```


Lab 6.2
Area and Perimeter

◆Step 3.檢查能否建構出三角形：

```
if(((in1 + in2 <= in3) || (in1 + in3 <= in2) || (in3 + in2 <= in1)) ||
   ((in1 - in2 >= in3) || (in1 - in3 >= in2) || (in2 - in1 >= in3)) ||
   (in3 - in1 >= in2) || ( in3 - in2 >= in1) || ( in2 - in3 >= in1)){
```

Lab 6.3
Grades Distribution Simulator

- In this problem, students are asked to use a random number generator to simulate the test grade distribution.
- The program takes a input to decide the number of testcases should be generated in this test simulation, and use random number generator to generate corresponding number of numerical grades.
- Each numerical grade is an integer with range[1,100].
- After that, the program will convert each numerical grade to letter grade based on the table shown in the next page.
- Then, the program will calculate the number of occurrence of each letter grade and print them out.
- Finally, the mean value and the STD deviation based on the numerical grade are calculated and printed. Note that the program rounds both of the mean value and the STD deviation to the second decimal place.
- You need to use time(0) as the seed of the random generator.

Lab 6.3
Grades Distribution Simulator

- The program converts the numerical grades to the letter grades as the table shown below.

Percentile	Letter Grade
90-100	A
80-89	B
70-79	C
60-69	D
0-59	F

Lab 6.3
Grades Distribution Simulator

- Input:
  - One **integer whose range is [1, 2^{31}-1] for deciding the number of sample.**
- Output:
  - The number of occurrence of each letter grade.
  - The mean value and the STD deviation based on the numerical grades.
  - Round both of the mean value and the STD deviation to the second decimal place.

Lab 6.3
Grades Distribution Simulator

- Hints: How to calculate the STD deviation

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2}$$
$$= \sqrt{\frac{1}{N}\left(\sum_{i=1}^{N} X_i^2\right) - \frac{1}{N}N\mu^2}$$
$$= \sqrt{\frac{\sum_{i=1}^{N} X_i^2}{N} - \mu^2}$$

Lab 6.3
Grades Distribution Simulator

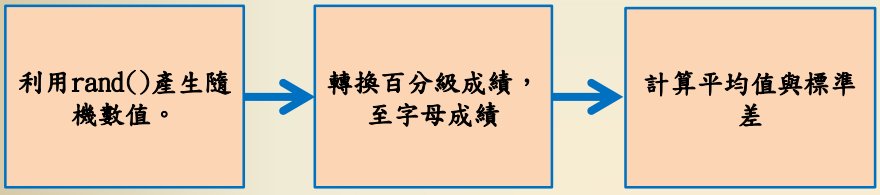
◆The screen dialog should appear as follows:

```
Enter # of sample: 5000
A: 510
B: 479
C: 491
D: 489
F: 3031
Mean: 49.49
STD Deviation: 28.69

Process returned 0 (0x0)   execution time : 5.808 s
Press any key to continue.
```


Lab 6.3
Grades Distribution Simulator

◆思考方向：



Lab 6.3
Grades Distribution Simulator

◆Step 1.利用rand()產生隨機數值：

```
14      srand(time(0));
15      num = 1 + rand() % 100;
```


Lab 6.3
Grades Distribution Simulator

◆Step 2.轉換百分級成績，至字母成績：

```
if (num >= 90 && num <= 100){
    cnt2++;
}
else if (num >= 80 && num < 90){
    cnt3++;
}
else if (num >= 70 && num < 80){
    cnt4++;
}
else if (num >= 60 && num < 70){
    cnt5++;
}
else if (num < 60){
    cnt6++;
}
```

Lab 6.3
Grades Distribution Simulator

◆Step 3.計算平均值與標準差：

```
59      //use total to get mean
60      double mean = (double)total / (double)sample;
61      cout << setprecision(2) << fixed;
62      roundTo2th(mean);
63      cout << "\nMean: " << mean << endl;
64
65      //use variance to get std deviation
66      variance = (variance/(double)sample) - pow(mean, 2);
67      double stdDev = sqrt(variance);
68      roundTo2th(stdDev);
69      cout << "STD Deviation: " << stdDev << endl;
70
```